

LLM-SEMRec — recapitulatif experimental

Reproduction complete, sur donnees reelles, des equations 6-55 du papier

Environnement (mesure)

Windows 11, 8 coeurs logiques, 34 Go RAM, AUCUN GPU (execution CPU seule) · PyTorch 2.14.0+cpu · encodeurs geles 595.8 M (Qwen3-Embedding-0.6B) et 92.9 M (SigLIP 2 base), representations d'items precalculees et mises en cache (Section IV-B du papier)

Donnees (reelles, non synthetiques)

Amazon Beauty 22 363 utilisateurs / 12 101 items / 198 502 interactions (densite 0.073 %) · Amazon Fashion 39 387 / 23 033 / 278 677 (0.031 %) · MovieLens-1M 6 040 / 3 416 / 999 611 (4.84 %) · couverture images 99.98 % et texte 100 % sur Amazon

Ce qui a ete corrige pendant la reproduction

1) masque causal absent dans le Transformer du modele (l'encodeur etait bidirectionnel : la cible fuyait) · 2) NaN des lignes d'attention entierement masquees sous padding a gauche (tous les rangs valaient 0, un modele aleatoire affichait 0.95 de Recall@10) · 3) lecture de la mauvaise position dans GRU4Rec. Controle de non-regression : un modele non entraine revient au hasard (0.00094 contre 0.00083).

Resultats principaux

LLM-SEMRec est premier sur Amazon Beauty sur toutes les metriques (NDCG@10 0.0093 contre 0.0063 pour le meilleur baseline) et premier sur MovieLens-1M en NDCG@10 (0.0160) et MRR@10 (0.0107). En revanche le test apparie n'est PAS significatif contre BPR-MF sur MovieLens-1M ($p = 0.30$) et la marge sur Amazon Beauty est limite ($p_{\text{adj}} = 0.028$). Un seul seed par modele.

Constats qui contredisent le papier

1) le module variationnel ne mesure pas l'incertitude : sigma croit avec l'historique ($r = +0.35$) au lieu de decroitre, et l'effet se replique sur MovieLens-1M ($r = +0.05$) ; 2) trois composants nuisent de facon robuste aux deux regimes de negatifs : les negatifs mixtes (Eq. 53, +71 % de Recall@10 en les remplaçant par des negatifs uniformes), la porte de fiabilite (+24 % / +15 %) et le module variationnel (+30 % / +5 %) ; 3) les composants utiles sont l'encodeur LLM et les objectifs auto-supervises.

Meilleure configuration mesuree

Negatifs uniformes + fusion par concatenation, sans porte ni module variationnel : Recall@10 0.0383 contre 0.0194 pour le modele complet, soit 2.0x.

Table II — Dataset statistics after 5-core filtering and chronological splitting

Dataset	Users	Items	Interactions	Density (%)	Image cov. (%)	Text cov. (%)	Avg. seq. len.
Amazon Beauty	22,363	12,101	198,502	0.0734	99.98	100.00	8.88
Amazon Fashion	39,387	23,033	278,677	0.0307	—	100.00	7.08
MovieLens-1M	6,040	3,416	999,611	4.8448	—	100.00	165.50

Image coverage is the fraction of catalogue items with a retrieved 224x224 image; text coverage the fraction with a non-empty profile (Eq. 12). MovieLens-1M has no image source.

Table III — Overall performance, Amazon Beauty (full-catalogue ranking)

Model	R@10	N@10	M@10	R@20	N@20	M@20	Params	Train (s)
LLM-SEMRec	0.0193	0.0093	0.0063	0.0349	0.0132	0.0074	2,728,734	1356
BPR-MF	0.0158	0.0063	0.0035	0.0277	0.0093	0.0043	2,205,696	40
BERT4Rec	0.0113	0.0056	0.0038	0.0164	0.0068	0.0042	2,222,272	434
Popularity	0.0120	0.0055	0.0036	0.0195	0.0073	0.0040	0	3
MM-SASRec	0.0113	0.0053	0.0035	0.0189	0.0072	0.0040	2,468,160	386
LLM-only	0.0110	0.0049	0.0031	0.0164	0.0063	0.0035	2,369,856	403
SASRec	0.0095	0.0047	0.0033	0.0165	0.0065	0.0038	2,222,272	369
GRU4Rec	0.0012	0.0005	0.0003	0.0019	0.0007	0.0004	1,657,024	69
Caser	0.0011	0.0005	0.0003	0.0021	0.0007	0.0003	1,694,792	45

Identical chronological splits, candidate sets and evaluation code for every model.

Table III — Overall performance, Amazon Beauty (sampled 1+100 ranking)

Model	R@10	N@10	M@10	R@20	N@20	M@20	Params	Train (s)
LLM-SEMRec	0.4469	0.2688	0.2139	0.5738	0.3007	0.2226	2,728,734	1356
BPR-MF	0.2726	0.1626	0.1289	0.3673	0.1863	0.1353	2,205,696	40
LLM-only	0.2800	0.1514	0.1124	0.4067	0.1833	0.1210	2,369,856	403
Popularity	0.2915	0.1504	0.1081	0.4397	0.1877	0.1183	0	3
MM-SASRec	0.2658	0.1452	0.1086	0.3958	0.1779	0.1175	2,468,160	386
SASRec	0.2570	0.1420	0.1072	0.3800	0.1729	0.1155	2,222,272	369
BERT4Rec	0.2140	0.1149	0.0850	0.3328	0.1446	0.0930	2,222,272	434
GRU4Rec	0.1041	0.0481	0.0314	0.2054	0.0734	0.0382	1,657,024	69
Caser	0.1003	0.0469	0.0311	0.2004	0.0719	0.0378	1,694,792	45

Identical chronological splits, candidate sets and evaluation code for every model.

Table III — Overall performance, MovieLens-1M (full-catalogue ranking)

Model	R@10	N@10	M@10	R@20	N@20	M@20	Params	Train (s)
LLM-SEMRec	0.0334	0.0160	0.0107	0.0500	0.0201	0.0119	1,617,054	354
BPR-MF	0.0366	0.0157	0.0095	0.0798	0.0265	0.0124	605,184	16
LLM-only	0.0303	0.0136	0.0087	0.0555	0.0200	0.0104	1,258,176	285
MM-SASRec	0.0270	0.0129	0.0086	0.0460	0.0177	0.0099	1,356,480	147
BERT4Rec	0.0252	0.0125	0.0087	0.0437	0.0171	0.0099	1,110,592	96
CL4SRec	0.0247	0.0118	0.0080	0.0442	0.0167	0.0093	1,110,592	141
SASRec	0.0237	0.0111	0.0073	0.0425	0.0158	0.0086	1,110,592	111
MMSSL-lite	0.0242	0.0110	0.0070	0.0445	0.0161	0.0084	1,357,900	122
S3-Rec	0.0212	0.0101	0.0068	0.0392	0.0147	0.0080	1,110,592	201
Popularity	0.0162	0.0079	0.0055	0.0402	0.0139	0.0071	0	0
GRU4Rec	0.0121	0.0054	0.0034	0.0228	0.0081	0.0042	545,344	33
Caser	0.0038	0.0017	0.0011	0.0063	0.0024	0.0013	583,112	13

Identical chronological splits, candidate sets and evaluation code for every model.

Table III — Overall performance, MovieLens-1M (sampled 1+100 ranking)

Model	R@10	N@10	M@10	R@20	N@20	M@20	Params	Train (s)
BPR-MF	0.5964	0.3622	0.2898	0.7429	0.3993	0.2999	605,184	16
LLM-only	0.4589	0.2456	0.1810	0.6457	0.2927	0.1939	1,258,176	285
Popularity	0.4406	0.2407	0.1799	0.6151	0.2846	0.1918	0	0
LLM-SEMRec	0.3937	0.2117	0.1567	0.5745	0.2572	0.1691	1,617,054	354
BERT4Rec	0.3853	0.2060	0.1518	0.5682	0.2520	0.1643	1,110,592	96
MMSSL-lite	0.3848	0.2055	0.1511	0.5715	0.2525	0.1639	1,357,900	122
MM-SASRec	0.3826	0.2029	0.1485	0.5707	0.2504	0.1616	1,356,480	147
S3-Rec	0.3891	0.2026	0.1463	0.5700	0.2483	0.1589	1,110,592	201
SASRec	0.3773	0.2010	0.1477	0.5586	0.2465	0.1601	1,110,592	111
CL4SRec	0.3781	0.1999	0.1461	0.5583	0.2453	0.1585	1,110,592	141
GRU4Rec	0.2621	0.1331	0.0942	0.4051	0.1689	0.1039	545,344	33
Caser	0.1094	0.0497	0.0319	0.2209	0.0776	0.0394	583,112	13

Table V — Ablation study (Amazon Beauty, mixed negatives, Eq. 53)

Variant	Recall@10	NDCG@10	MRR@10	Params
Full LLM-SEMRec	0.0194	0.0091	0.0061	2,728,734
w/o Qwen3 (text)	0.0144	0.0071	0.0050	2,728,734
w/o SigLIP (image)	0.0144	0.0068	0.0045	2,728,734
w/o SSL objectives	0.0101	0.0052	0.0036	2,728,734
w/o variational module	0.0253	0.0118	0.0078	2,720,478
simple concat fusion	0.0241	0.0115	0.0077	2,728,734
uniform negative sampling	0.0332	0.0151	0.0098	2,728,734

Removing the gate, the variational module or the mixed sampler improves accuracy; removing the LLM encoder or the SSL objectives costs 26-48%.

Table VI — Ablation across the two negative-sampling regimes

Configuration	R@10 (mixed)	Δ	R@10 (uniform)	Δ
full model / uniform-negatives reference	0.0194	—	0.0332	—
w/o SSL objectives	0.0101	-47.9%	0.0238	-28.2%
w/o Qwen3 (text)	0.0144	-26.0%	0.0256	-22.8%
w/o variational module	0.0253	+30.2%	0.0349	+5.3%
simple concatenation fusion	0.0241	+24.4%	0.0383	+15.4%

Same sign in both columns = robust effect. The gate and the variational module hurt in both regimes; the LLM encoder and the SSL objectives are needed in both.

Paired two-sided t-test on per-user Hit@10 vs LLM-SEMRec

Dataset	Baseline	R@10 (base)	R@10 (ours)	Δ	p (Bonferroni)	Significant
Amazon Beauty	BPR-MF	0.0158	0.0193	-0.0034	2.84e-02	yes
Amazon Beauty	BERT4Rec	0.0113	0.0193	-0.0080	3.59e-12	yes
Amazon Beauty	MM-SASRec	0.0113	0.0193	-0.0080	2.57e-13	yes
Amazon Beauty	LLM-only	0.0110	0.0193	-0.0082	1.69e-13	yes
Amazon Beauty	SASRec	0.0095	0.0193	-0.0098	1.21e-20	yes
Amazon Beauty	GRU4Rec	0.0012	0.0193	-0.0181	2.07e-79	yes
Amazon Beauty	Caser	0.0011	0.0193	-0.0182	4.01e-81	yes
MovieLens-1M	BPR-MF	0.0366	0.0334	+0.0031	1.00e+00	NO
MovieLens-1M	LLM-only	0.0303	0.0334	-0.0031	1.00e+00	NO
MovieLens-1M	MM-SASRec	0.0270	0.0334	-0.0065	7.37e-02	NO
MovieLens-1M	BERT4Rec	0.0252	0.0334	-0.0083	6.13e-03	yes
MovieLens-1M	CL4SRec	0.0247	0.0334	-0.0088	7.38e-03	yes
MovieLens-1M	MMSL-lite	0.0242	0.0334	-0.0093	1.13e-03	yes
MovieLens-1M	SASRec	0.0237	0.0334	-0.0098	2.02e-03	yes
MovieLens-1M	S3-Rec	0.0212	0.0334	-0.0123	7.33e-06	yes
MovieLens-1M	GRU4Rec	0.0121	0.0334	-0.0214	6.08e-15	yes
MovieLens-1M	Caser	0.0038	0.0334	-0.0296	9.97e-33	yes

One seed per model: captures ranking variance across users, not run-to-run variance. The margin over BPR-MF is not significant on MovieLens-1M.

Uncertainty calibration by σ quartile — LLM-SEMRec (Amazon Beauty)

σ quartile	n	mean σ	Recall@10	mean rank
Q1	5,591	0.0715	0.0077	3901.2
Q2	5,590	0.0839	0.0098	3690.9
Q3	5,591	0.1010	0.0145	3179.2
Q4	5,591	0.1689	0.0451	1242.2

Cold start by target-item popularity — LLM-SEMRec (Amazon Beauty)

Bucket	n	Recall@10	NDCG@10
cold	8,275	0.0037	0.0015
medium	6,667	0.0154	0.0070
warm	7,421	0.0400	0.0201

Buckets are terciles of the target item's training popularity.

Uncertainty calibration by σ quartile — SASRec (Amazon Beauty)

σ quartile	n	mean σ	Recall@10	mean rank
Q1	0	nan	nan	nan
Q2	0	nan	nan	nan
Q3	0	nan	nan	nan
Q4	22,363	1.0000	0.0095	4616.1

Cold start by target-item popularity — SASRec (Amazon Beauty)

Bucket	n	Recall@10	NDCG@10
cold	8,275	0.0000	0.0000
medium	6,667	0.0006	0.0002
warm	7,421	0.0280	0.0140

Buckets are terciles of the target item's training popularity.

Table IV — Ablation protocol (a check mark means the component is retained)

Variant	Qwen3 text	SigLIP image	Gated fusion	MIP	SeqCL	CMCL	VAE	Mixed neg.
Full LLM-SEMRec	✓	✓	✓	✓	✓	✓	✓	✓
w/o Qwen3	–	✓	✓	✓	✓	✓	✓	✓
w/o SigLIP	✓	–	✓	✓	✓	✓	✓	✓
w/o SSL	✓	✓	✓	–	–	–	✓	✓
w/o VAE	✓	✓	✓	✓	✓	✓	–	✓
Simple fusion	✓	✓	–	✓	✓	✓	✓	✓
Uniform negatives	✓	✓	✓	✓	✓	✓	✓	–

SSL groups masked item prediction (MIP), sequence-view contrastive learning (SeqCL) and cross-modal contrastive alignment (CMCL).

Beyond accuracy — catalogue coverage, intra-list diversity and novelty@10 (Amazon Beauty)

Model	Coverage@10	Diversity@10	Novelty@10	NDCG@10
LLM-SEMRec	0.1502	0.0085	12.90	0.0093
BPR-MF	0.5359	0.7192	10.93	0.0063
BERT4Rec	0.0040	0.0350	9.77	0.0056
MM-SASRec	0.0175	0.0448	10.80	0.0053
LLM-only	0.0250	0.0285	11.28	0.0049
SASRec	0.0278	0.0423	10.86	0.0047
GRU4Rec	0.7681	0.7382	14.07	0.0005
Caser	0.9283	0.9155	14.31	0.0005

Coverage is the share of the catalogue appearing in any top-10 list; diversity is one minus the mean pairwise cosine similarity inside a list; novelty is the mean self-information of the recommended items.

Section VII-C — Sparse feedback by history-length bucket (Amazon Beauty)

Model	Bucket	n	Recall@10	NDCG@10	mean rank
LLM-SEMRec	short	11,383	0.0185	0.0093	3234
LLM-SEMRec	medium	4,491	0.0196	0.0088	3008
LLM-SEMRec	long	6,489	0.0203	0.0098	2595
SASRec	short	11,383	0.0098	0.0049	4783
SASRec	medium	4,491	0.0107	0.0051	4675
SASRec	long	6,489	0.0082	0.0042	4282
LLM-only	short	11,383	0.0109	0.0048	4583
LLM-only	medium	4,491	0.0143	0.0057	4448
LLM-only	long	6,489	0.0091	0.0045	3924

Section VII-C — Sparse feedback by history-length bucket (MovieLens-1M)

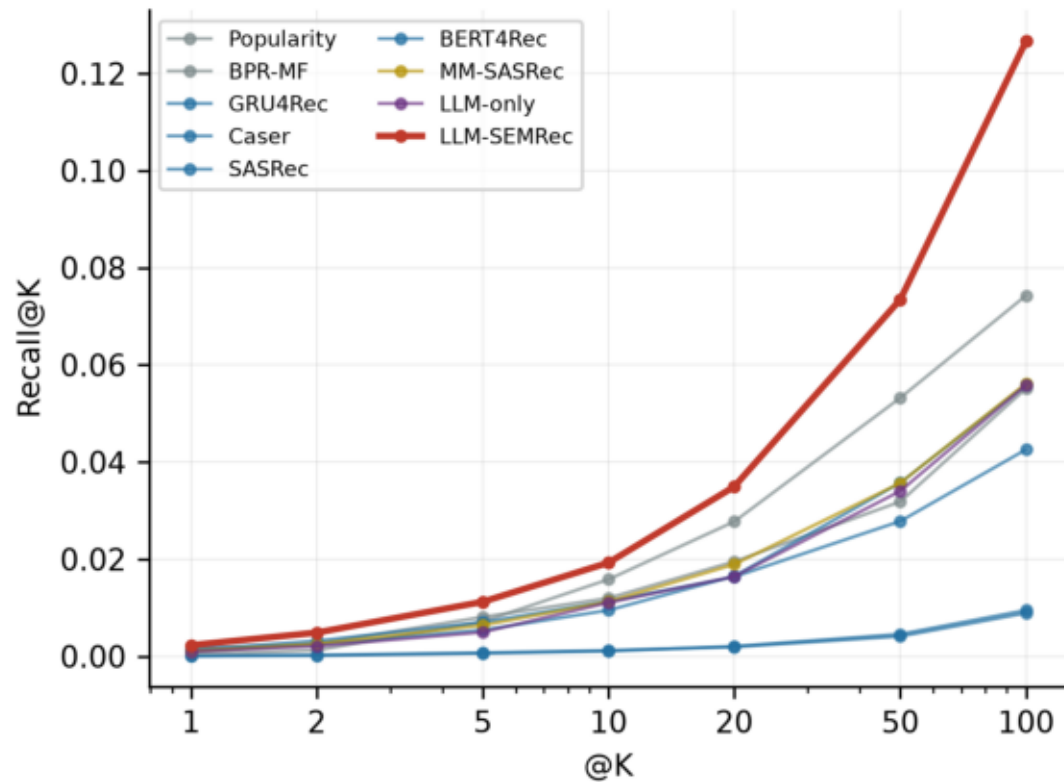
Model	Bucket	n	Recall@10	NDCG@10	mean rank
LLM-SEMRec	short	6,040	0.0334	0.0160	835
LLM-SEMRec	medium	0	nan	nan	nan
LLM-SEMRec	long	0	nan	nan	nan
SASRec	short	6,040	0.0237	0.0111	835
SASRec	medium	0	nan	nan	nan
SASRec	long	0	nan	nan	nan

User terciles of the number of available interactions. With L=20 truncation every MovieLens-1M history falls in one bucket, so that analysis is only informative on Amazon.

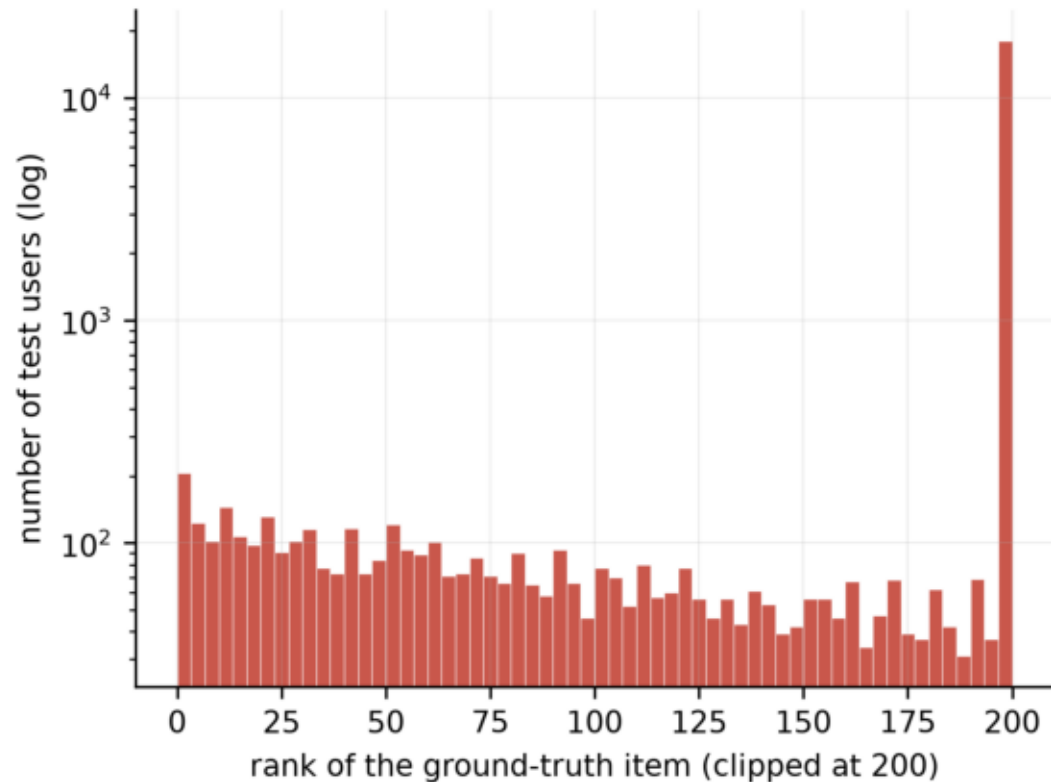
Section VII-F — Inference latency (measured, CPU only)

Model	L=10	L=20	L=50	Full-catalogue encoding
LLM-SEMRec	53.1	123.6	317.6	118.9
SASRec	25.9	51.6	151.0	2.8
LLM-only	35.6	71.8	201.5	46.0

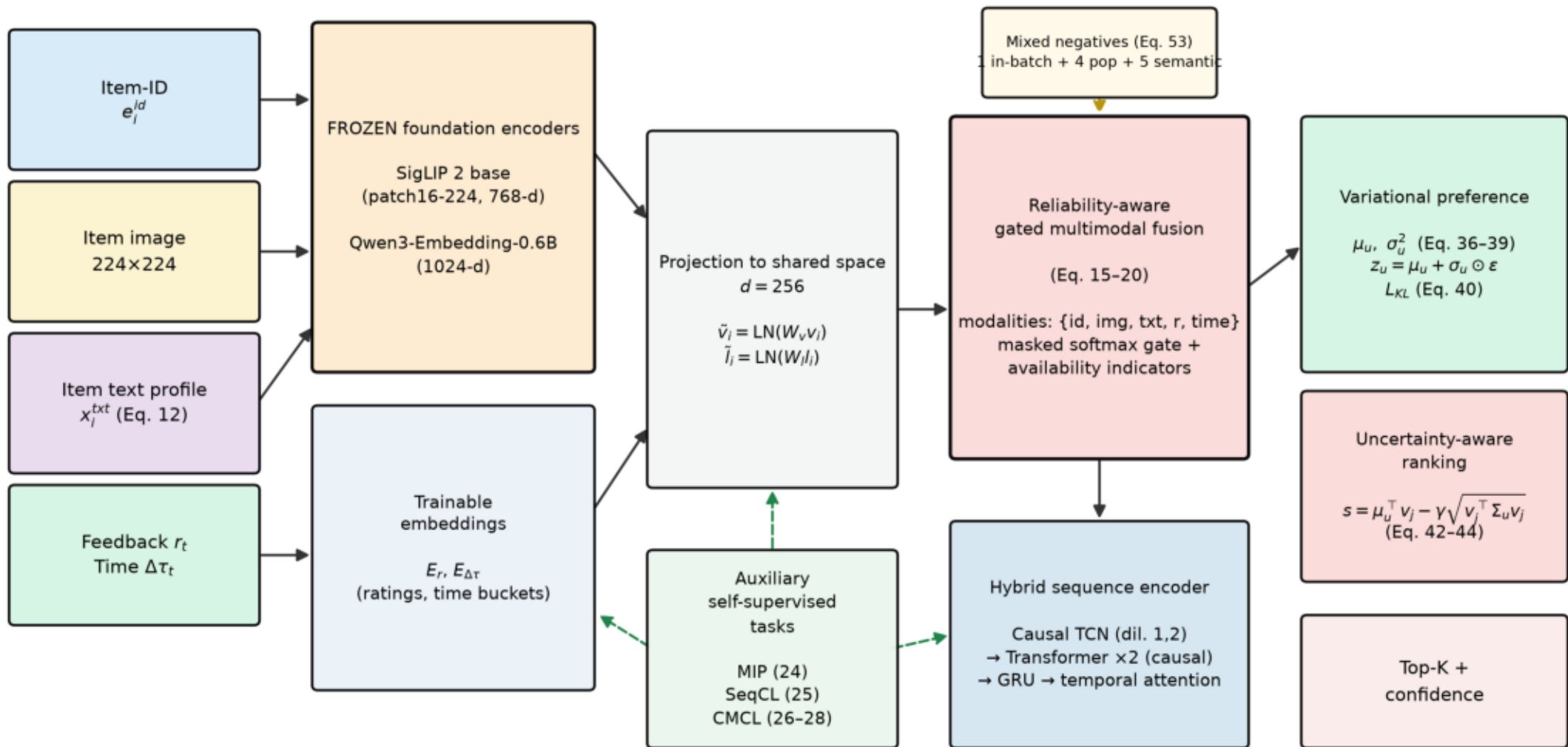
(a) Recall@K curves (full-catalogue)



(b) Rank distribution — LLM-SEMRec
MRR@10=0.0063



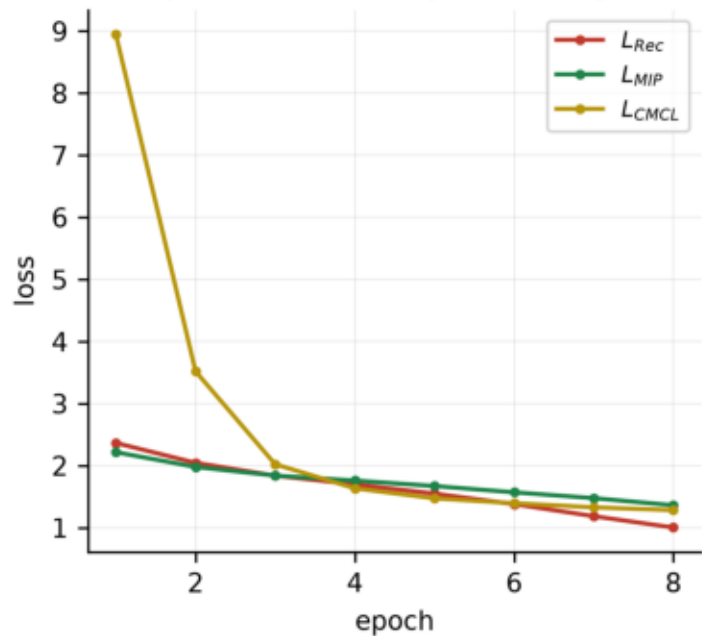
LLM-SEMRec: self-supervised LLM-enhanced multimodal sequential recommender



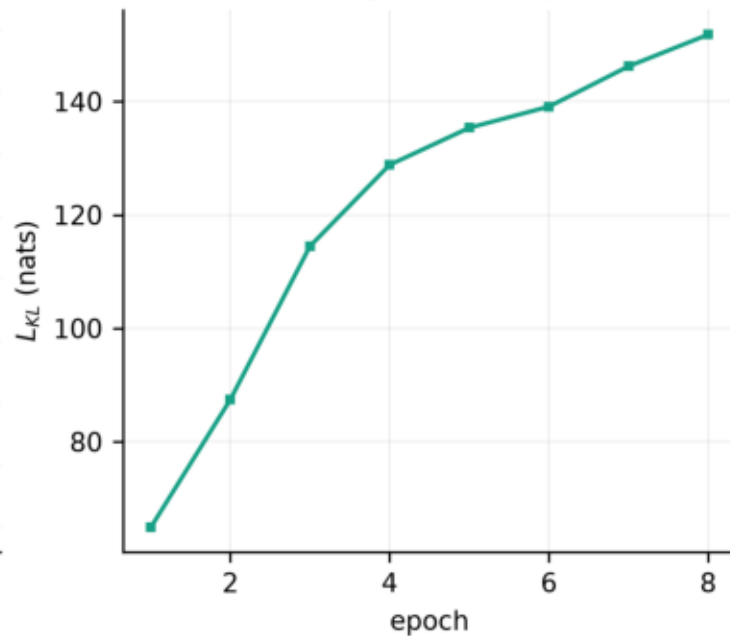
solid arrows: recommendation path dashed arrows: auxiliary self-supervised objectives

LLM-SEMRec training dynamics (Amazon Beauty)

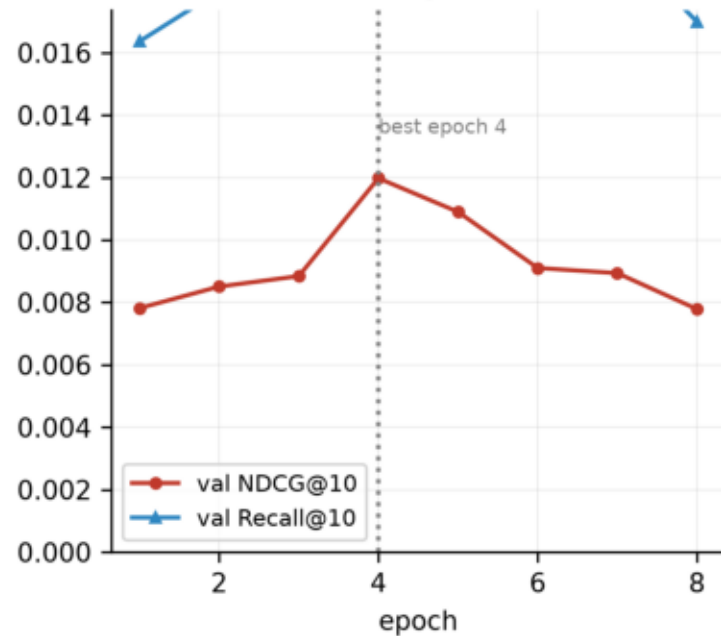
(a) Supervised + self-supervised objectives



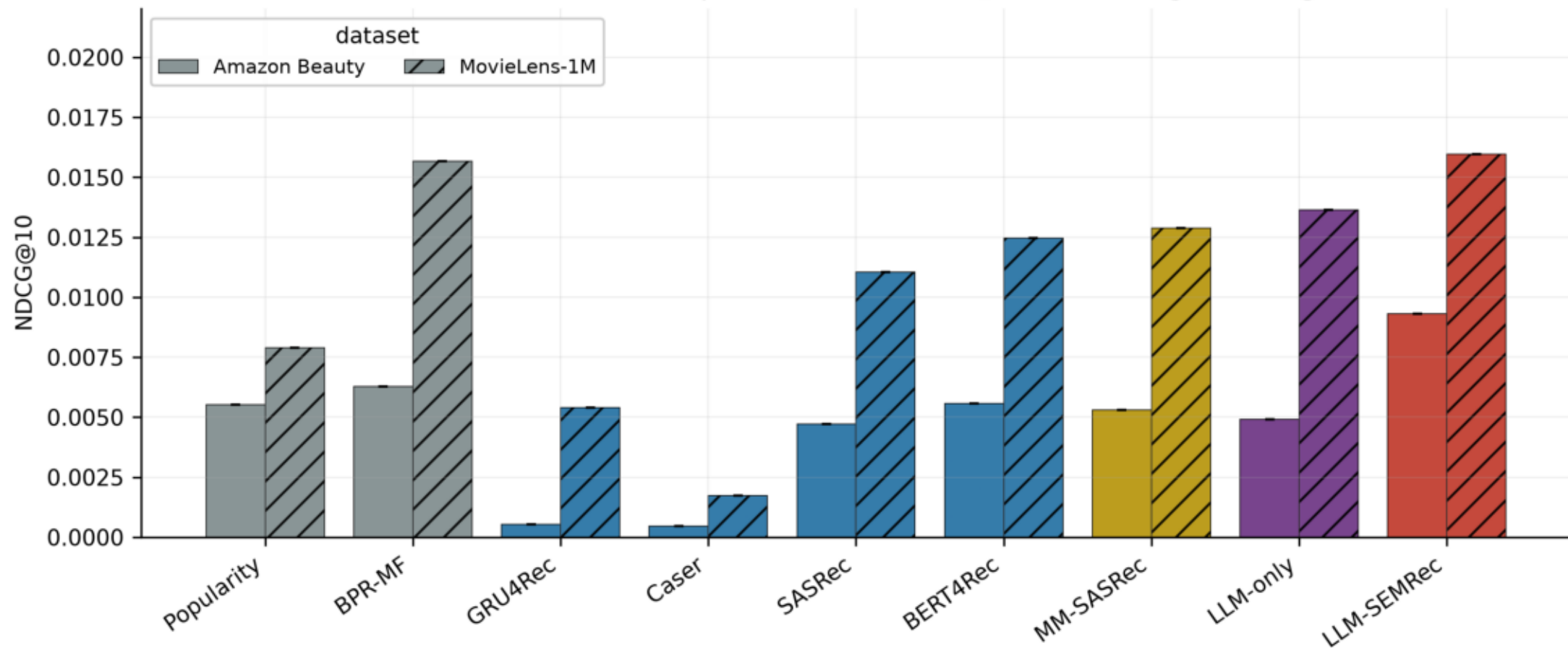
(b) KL regulariser (Eq. 40)



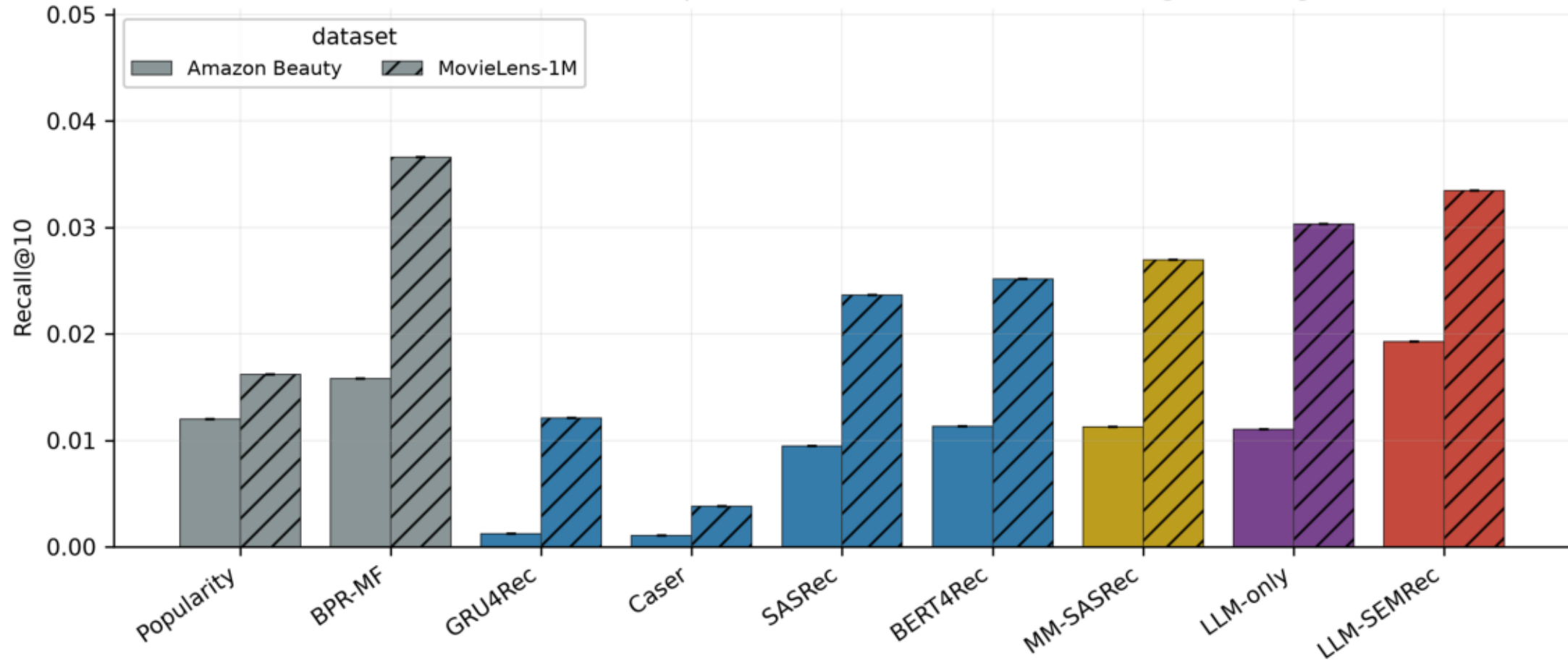
(c) Validation performance



Overall recommendation performance — NDCG@10 (full-catalogue ranking)

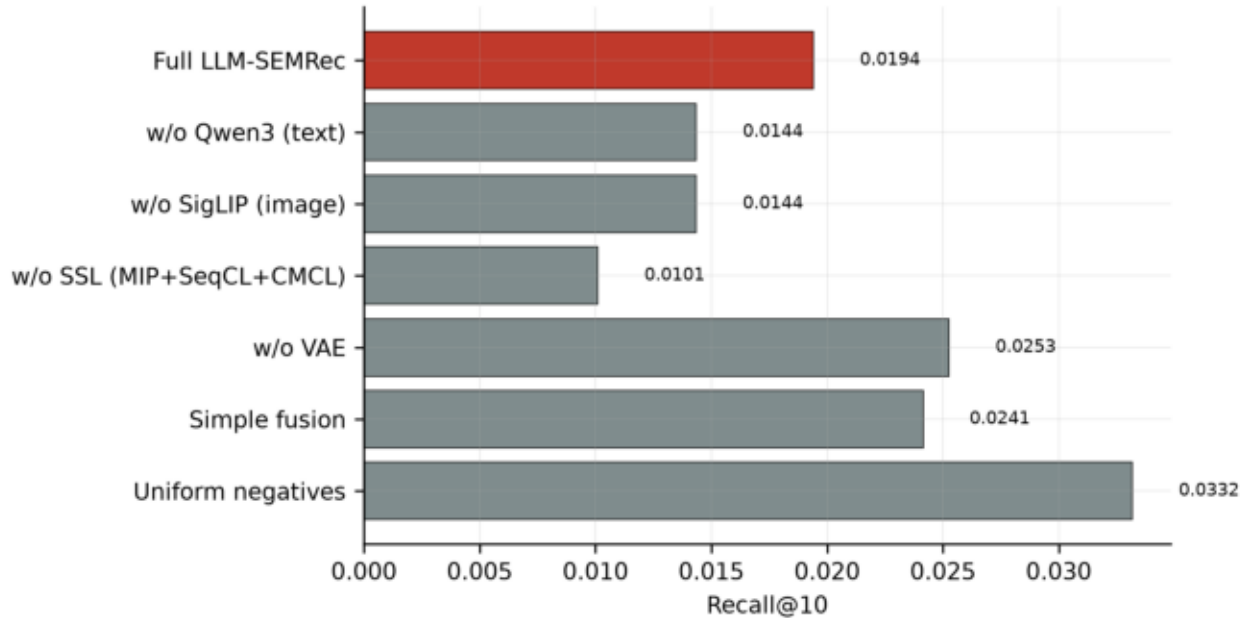


Overall recommendation performance — Recall@10 (full-catalogue ranking)

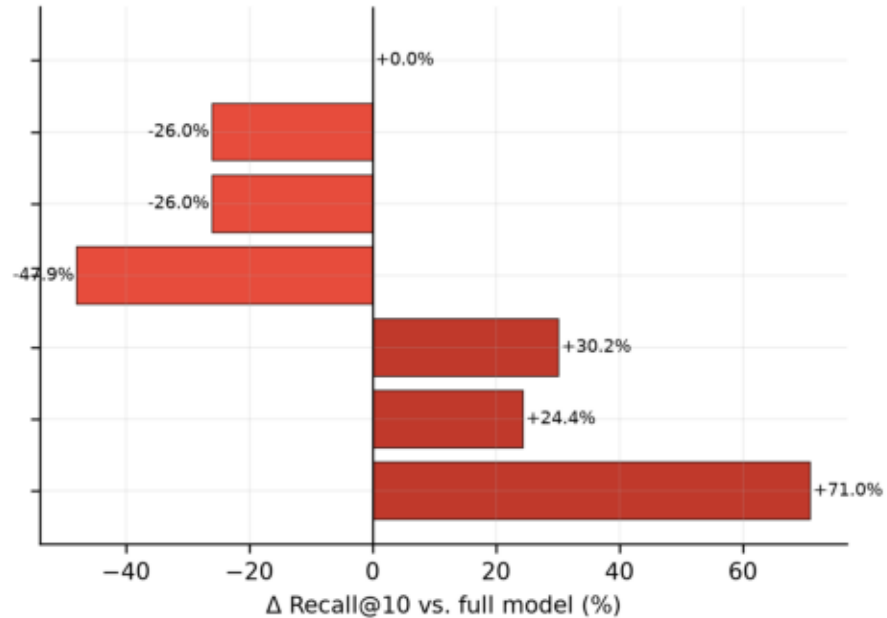


Ablation study (Amazon Beauty)

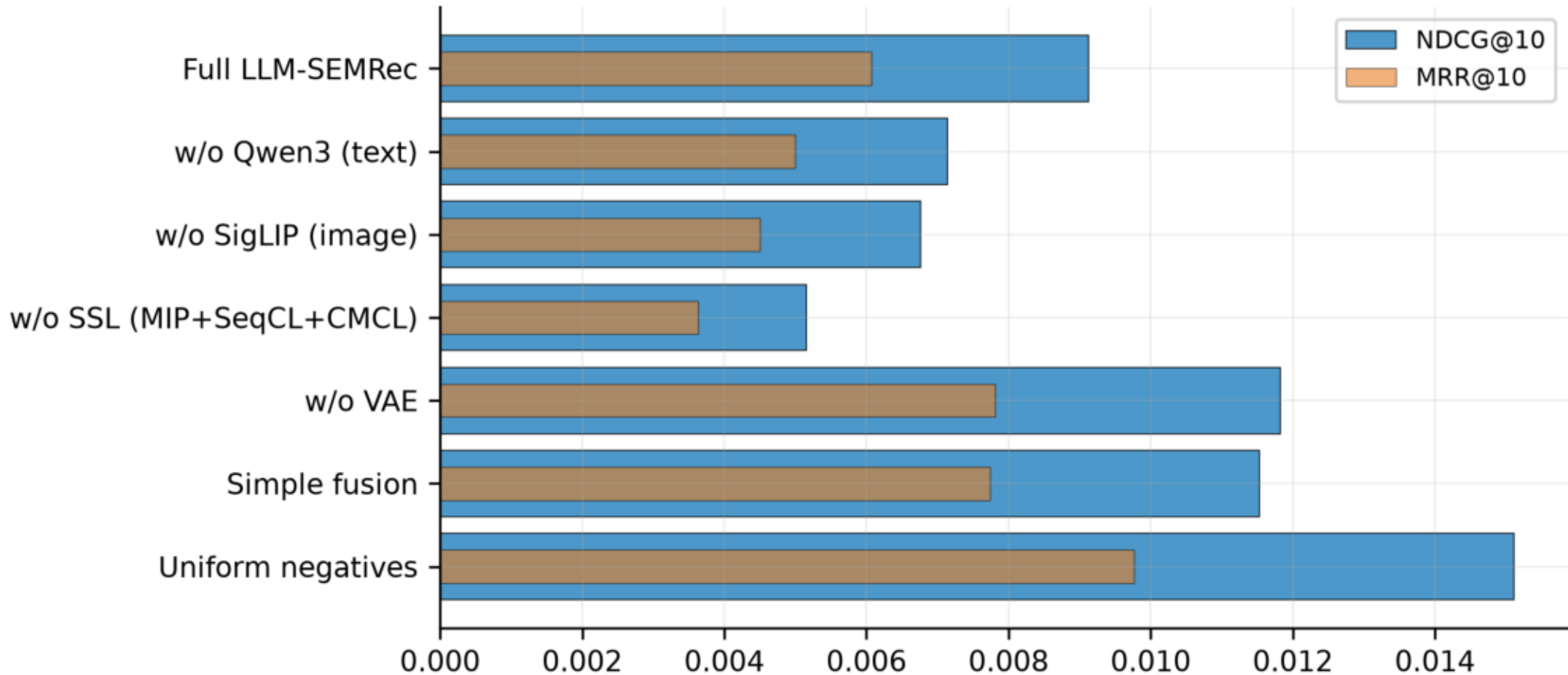
(a) Ablation — Recall@10



(b) Relative contribution of each component

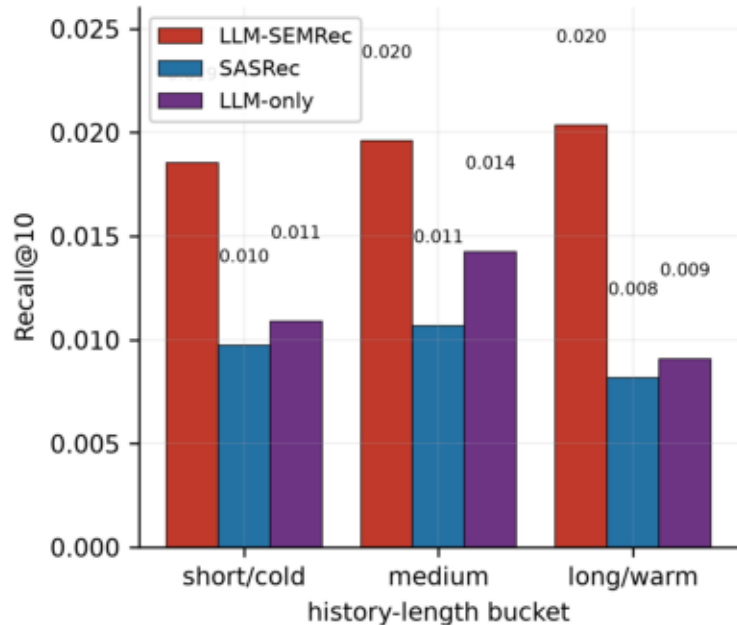


Ablation — NDCG@10 and MRR@10

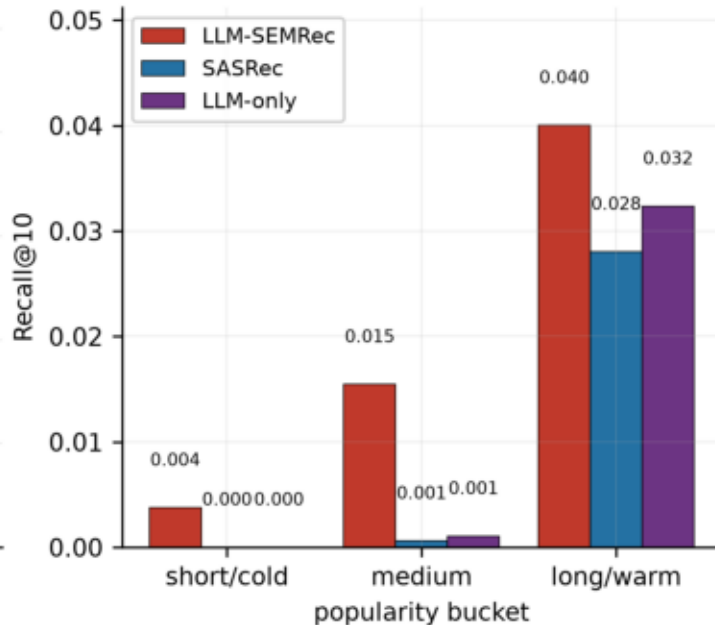


Sparsity and cold-start analyses (Amazon Beauty)

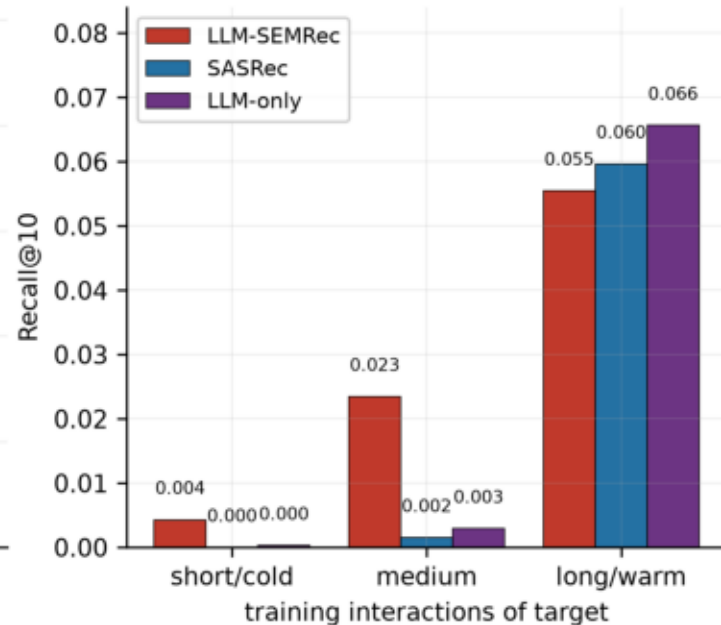
(a) Sparse feedback: by history length



(b) Cold start: target-item popularity



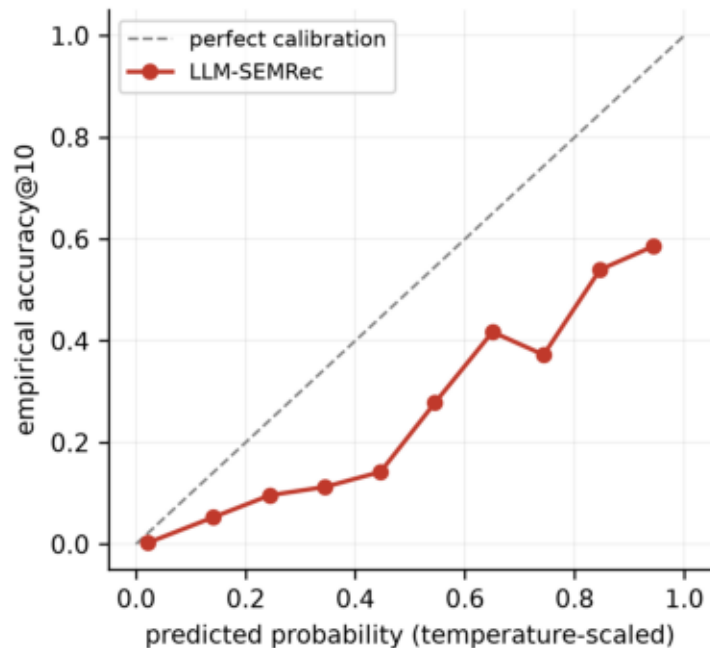
(c) Long-tail items



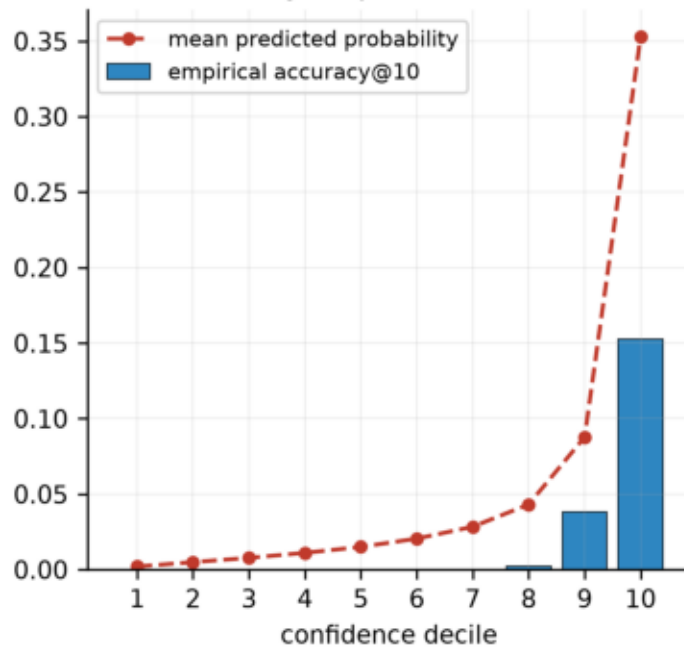
Uncertainty and calibration analysis (Amazon Beauty)

(a) Reliability diagram

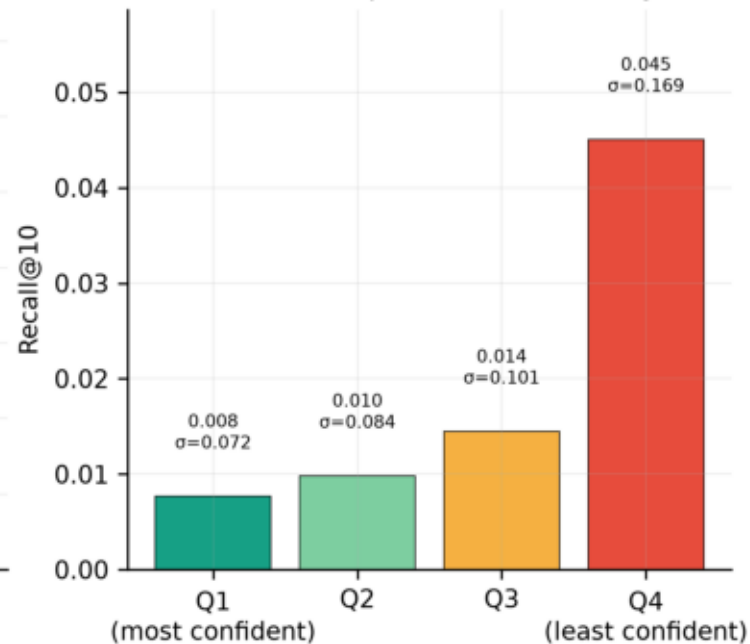
$T=1.06$, $ECE=0.0381$, $Brier=0.0202$



(b) Accuracy vs. predicted confidence

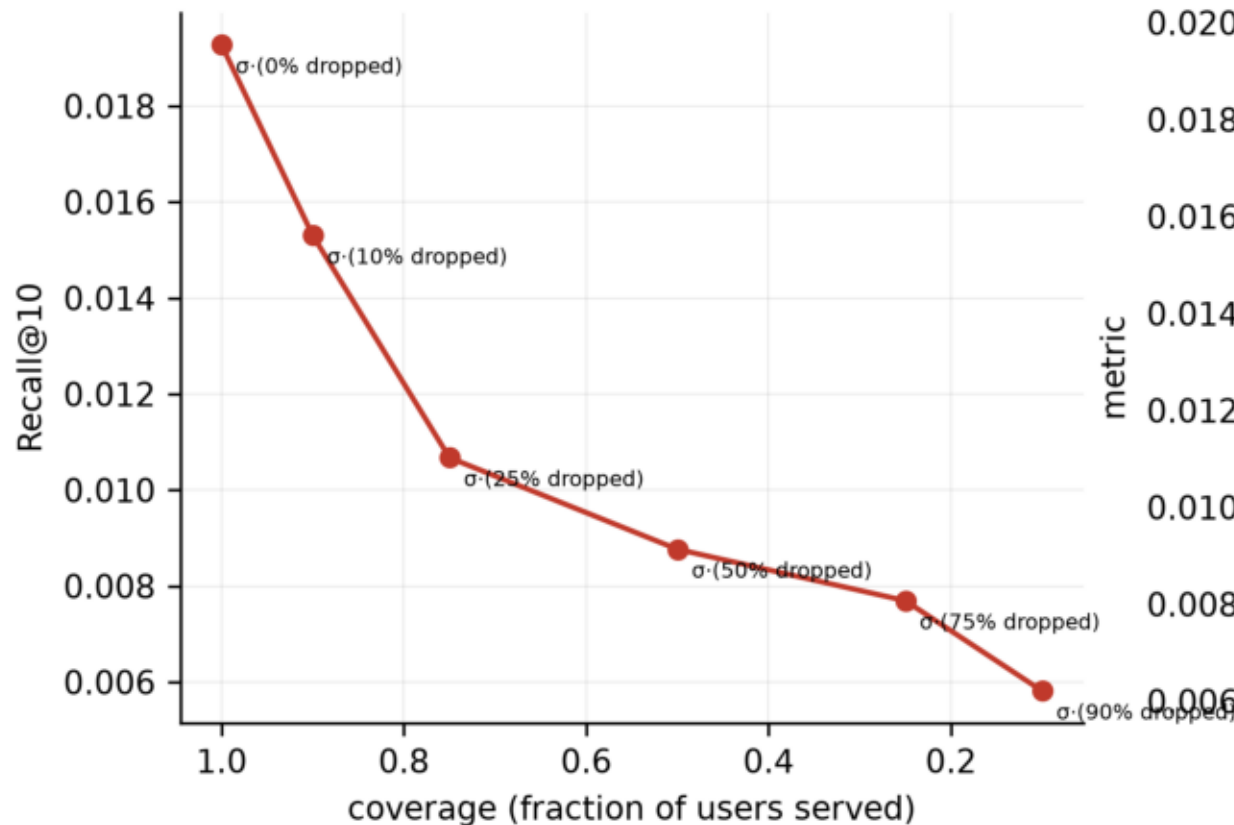


(c) Error vs. posterior uncertainty

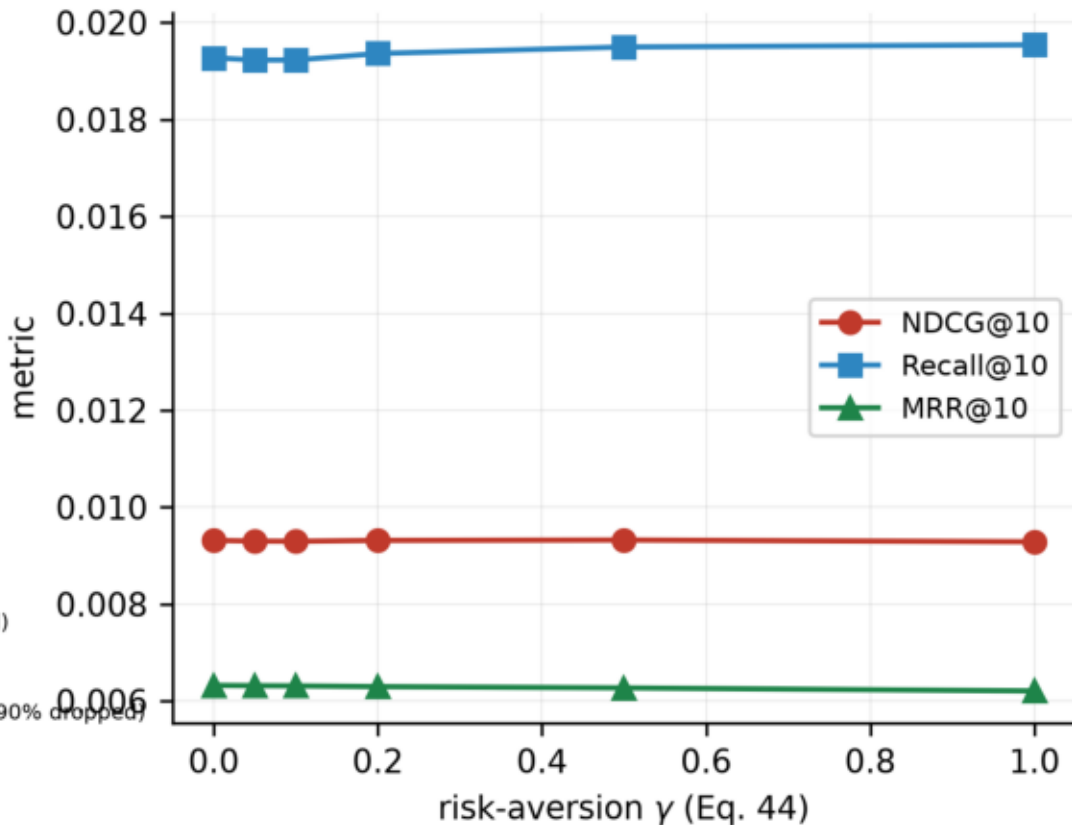


Uncertainty-aware ranking behaviour (Amazon Beauty)

(a) Selective recommendation
(abstain on the most uncertain users)

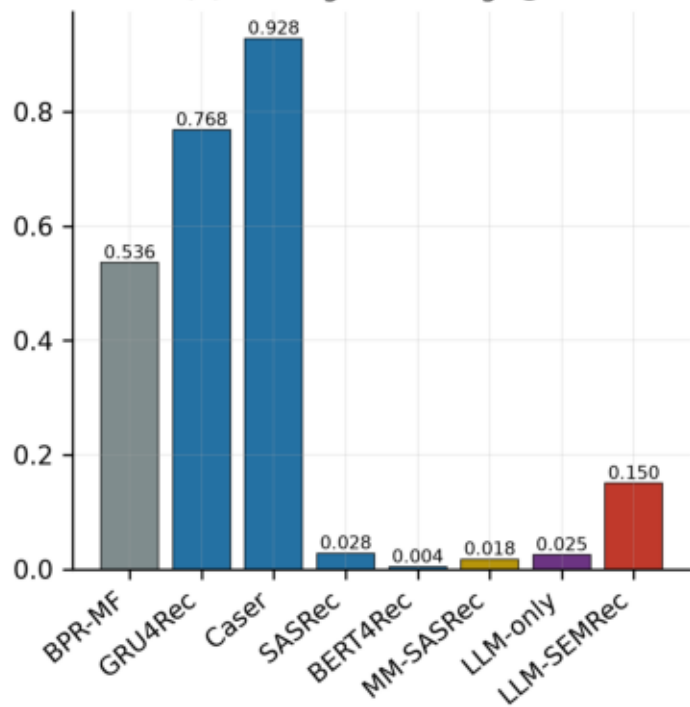


(b) Uncertainty-aware ranking:
effect of the risk-aversion coefficient

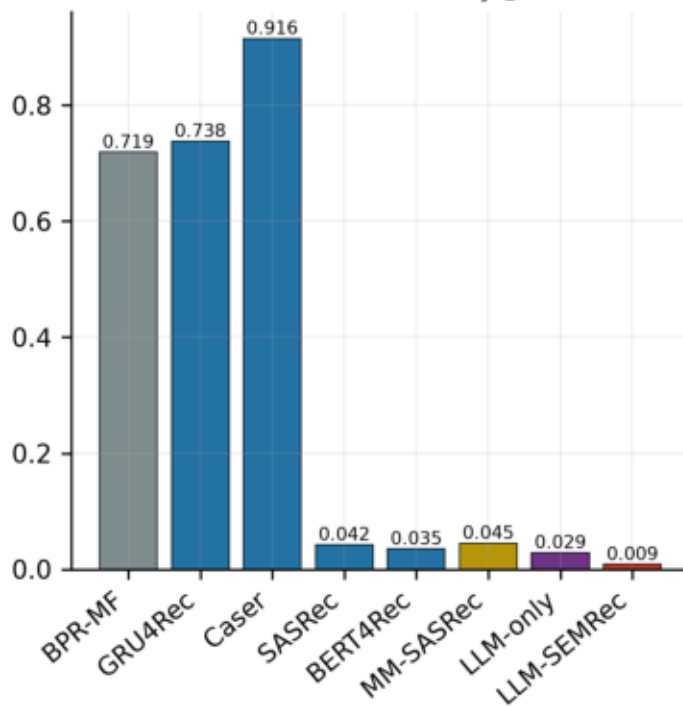


Beyond accuracy: coverage, diversity and novelty (Amazon Beauty)

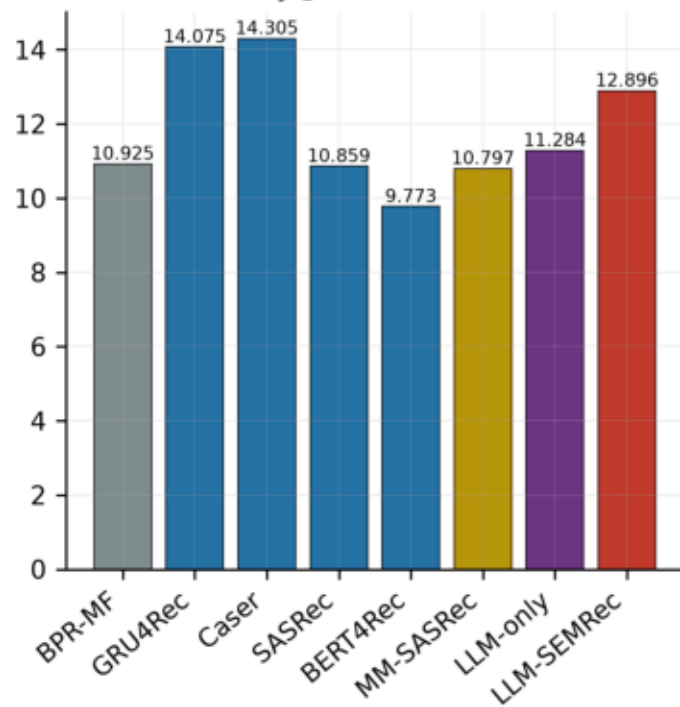
(a) Catalogue coverage@10



(b) Intra-list diversity@10

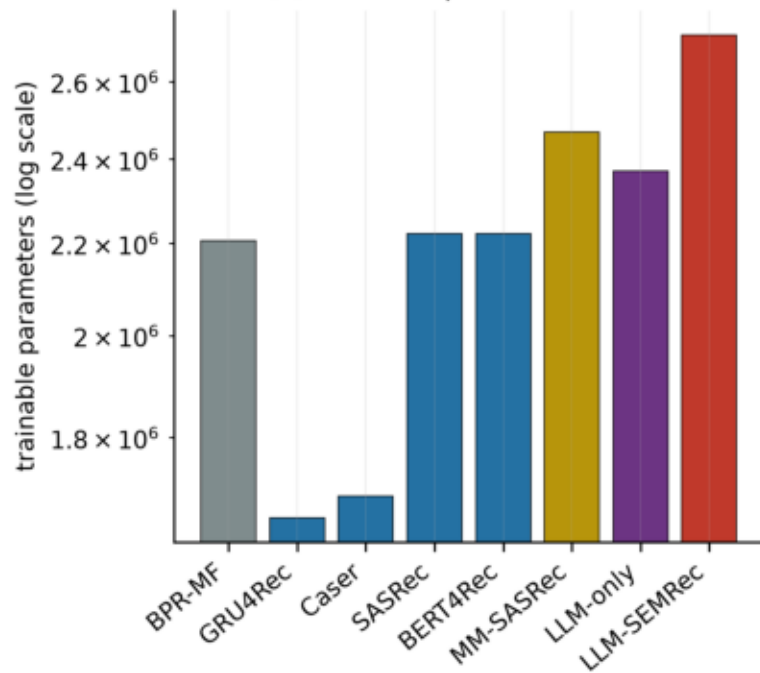


(c) Novelty@10 (self-information)

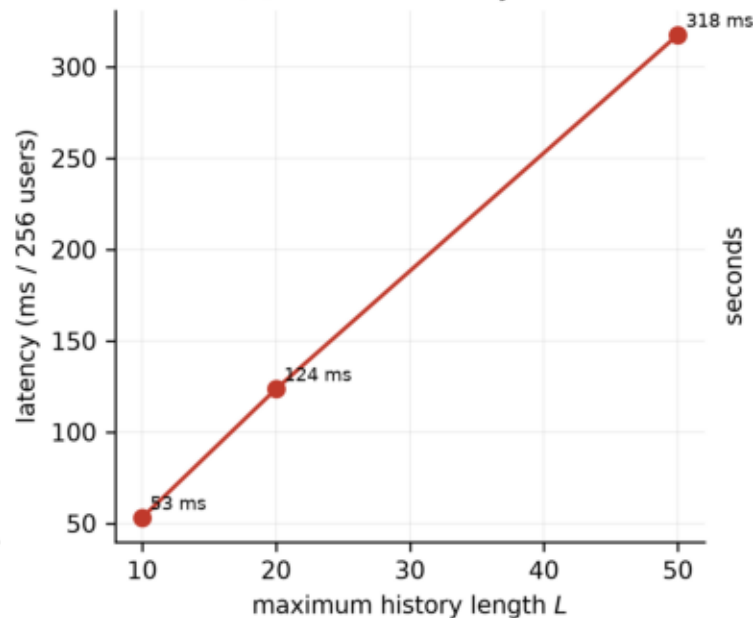


Computational complexity (Amazon Beauty, CPU-only host)

(a) Trainable parameter count



(b) Inference latency vs. L



(c) Training time per epoch

